

Week 6 – Demo 6: Data Visualization, Context & Storytelling

Dataset: MTCARS & CEREALS

Lab Questions (MTCARS Dataset)

1. Load the **MTCARS** dataset and display its basic structure (rows, columns, data types).
2. Compute summary statistics (mean, median, min, max) for:
 - Miles per gallon (mpg)
 - Horsepower (hp)
 - Weight (wt)
3. Identify the relationship between engine cylinders (cyl) and fuel efficiency (mpg) using an appropriate visualization.
4. Create a scatter plot between horsepower (hp) and mpg.
 - Add labels and a meaningful title.
5. Compare manual vs automatic transmission cars in terms of mpg using a box plot.
6. Identify which car model gives the best balance of power and fuel efficiency. Justify using a visualization.
7. Write a short data story (3–4 lines) summarizing insights derived from the plots.

Additional questions if required

1. Load the **CEREALS** dataset and inspect missing values.
2. Calculate average calories, sugar, and fiber content.
3. Create a bar chart showing top 5 cereals with highest sugar content.
4. Visualize the relationship between calories and rating.
5. Compare cold vs hot cereals based on calorie distribution.
6. Identify healthier cereals using at least two nutritional parameters.
7. Summarize findings using one key visualization and explanation.

Week 7 – Demo 7: Plotting using Matplotlib & Seaborn

Dataset: IPL

Lab Questions

1. Load the **IPL dataset** and display first 10 records.
2. Find the total number of:
 - Matches
 - Teams
 - Seasons
3. Plot the number of matches played per season using Matplotlib.
4. Use Seaborn to visualize:
 - Top 5 teams with most wins
5. Create a bar plot showing top 10 run scorers.
6. Plot a pie chart representing match wins by venue.
7. Compare home vs away wins using an appropriate plot.
8. Create a visualization that tells a clear IPL performance story (your choice).
9. Add annotations, legends, and titles to improve readability.

Week 8 – Demo 8: Data Aggregation & Group Operations

Dataset: NORTHWIND

Lab Questions

1. Load the NORTHWIND dataset and explore all available tables.
2. Calculate:
 - Total sales per product
 - Total sales per category
3. Identify the top 5 customers by revenue.
4. Group data by country and compute total orders.
5. Find the monthly sales trend using aggregation.
6. Perform a groupby operation to calculate average order value per employee.
7. Create a bar chart showing category-wise sales.
8. Visualize country-wise revenue contribution.

9. Interpret results and write 2 business insights based on aggregation.

Week 9 – Demo 9: String Manipulation & Data Wrangling

Dataset: SENTIMENT

Lab Questions

1. Load the SENTIMENT dataset and inspect text columns.
2. Perform basic string operations:
 - Convert text to lowercase
 - Remove punctuation
3. Count the number of positive, negative, and neutral sentiments.
4. Find the most frequently occurring words in positive reviews.
5. Remove stopwords and display cleaned text.
6. Create a word length distribution plot.
7. Generate a bar chart showing sentiment count.
8. Identify top keywords influencing negative sentiment.
9. Write a short summary explaining how string processing helped sentiment analysis.